Worksheet 32 · Chord Theorems IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Use the intersecting chords theorem • Use the tangent–secant theorem • Solve problems involving chords and tangents		
Grade: / 35	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Chord Theorems*

Question 1 (2 marks)

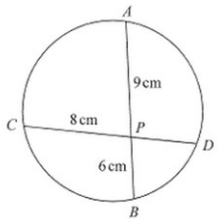


Diagram **NOT** accurately drawn

APB and *CPD* are chords of a circle.

$AP = 9$ cm $PB = 6$ cm $CP = 8$ cm

Calculate the length of *PD*.

..... cm

Question 2 (3 marks)

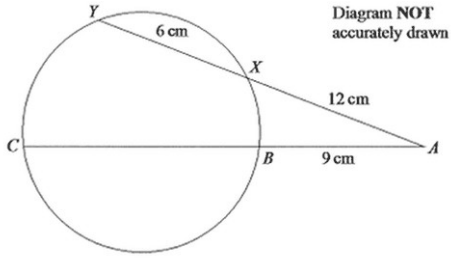


Diagram **NOT** accurately drawn

The points *B*, *C*, *Y* and *X* lie on a circle.
AXY and *ABC* are straight lines.
$AX = 12$ cm $XY = 6$ cm $AB = 9$ cm

Calculate the length of *BC*.

.....cm

Question 3 (7 marks)

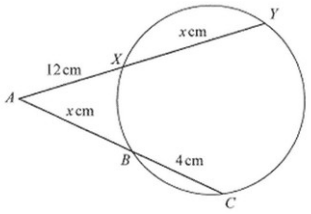


Diagram **NOT** accurately drawn

The points *B*, *C*, *Y* and *X* lie on a circle.

AXY and *ABC* are straight lines.

$AX = 12$ cm $XY = x$ cm $AB = x$ cm $BC = 4$ cm

(a) Show that $x^2 - 8x - 144 = 0$

(3)

(b) Find the length of *AC*.
Show your working clearly.
Give your answer correct to 3 significant figures.

..... cm

(4)

Question 4 (6 marks)

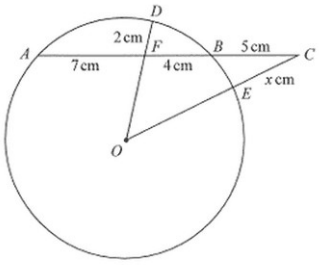


Diagram **NOT** accurately drawn

A, *D*, *B* and *E* are points on a circle, centre *O*.
AFBC, *OEC* and *OFD* are straight lines.

$AF = 7$ cm, $FB = 4$ cm, $BC = 5$ cm, $FD = 2$ cm and $CE = x$ cm.

Work out the value of *x*.
Show your working clearly.

$x =$ .....

Question 5 (3 marks)

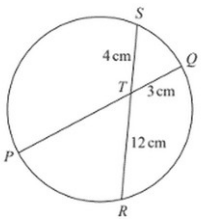


Diagram **NOT** accurately drawn

PTQ is a diameter of a circle.
RTS is a chord of the circle.

$TQ = 3$ cm $\quad ST = 4$ cm $\quad TR = 12$ cm

Calculate the radius of the circle.

..... cm

Question 6 (3 marks)

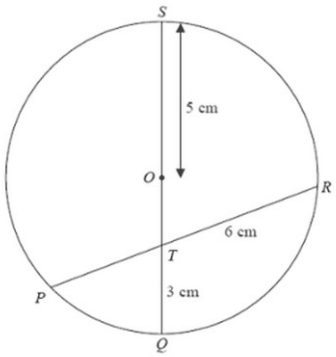


Diagram **NOT** accurately drawn

P, Q, R and S are points on a circle, centre O.
QS is a diameter of the circle.
QS and PR intersect at the point T.
$OS = 5$ cm, $QT = 3$ cm and $TR = 6$ cm.

Work out the length of PT.

.....cm

Question 7 (3 marks)

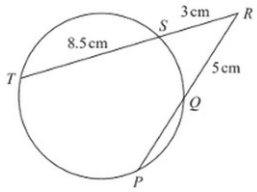


Diagram **NOT** accurately drawn

P, Q, S and T are points on a circle.
TSR and PQR are straight lines.

Work out the length of PQ.

..... cm

Question 8 (3 marks)

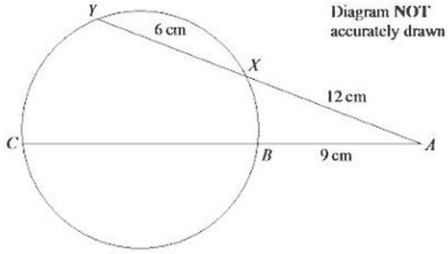


Diagram **NOT** accurately drawn

The points B, C, Y and X lie on a circle.
AXY and ABC are straight lines.
$AX = 12$ cm $\quad XY = 6$ cm $\quad AB = 9$ cm

Calculate the length of BC.

.....cm

Question 9 (5 marks)

AEC and BED are chords of a circle.

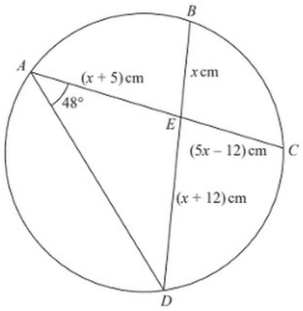


Diagram **NOT** accurately drawn

$AE = (x + 5)$ cm $\quad BE = x$ cm $\quad CE = (5x - 12)$ cm $\quad DE = (x + 12)$ cm

Angle $DAE = 48°$

Work out the size of angle ADE
Give your answer correct to one decimal place.

°